Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of Valladolid
Short name / acronym used in projects	UVa
Participant Identification Code (PIC), if available	
Website(s)	https://www.uva.es/export/sites/uva/
Country	SPAIN
Organisation type	X Academic ☐ Non-academic X Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	
Department(s), institute(s), research group(s) or unit(s) that would be involved	Group name: Biomarkers in Neurodegenerative, Metabolic, and Inflammatory Pathology Department of Cell Biology. Faculty of Medicine. University of Valladolid IOBA- University of Valladolid
Additional institutional resources relevant to collaboration	

Main contact person for this profile	Associate Prof. Dr. SALVADOR PASTOR IDOATE and Prof. RICARDO USATEGUI MARTÍN

Main contact email	Salvador.pastor@uva.es and ricardo.usategui@uva.es
Other contact person(s), if relevant	

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	X Retina
X Inherited retinal dystrophies (IRDs)	X Age-related ocular disease
X Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	X AI-assisted diagnosis / software / digital tools
X Clinical trials / translational validation	☐ Other (please specify):

Free-Keywords for internal searchable database (please provide up to 10)	retinal organoids; neuroinflammation; AI diagnostics; drug delivery; biomarkers; neuroprotection; retina; retinal detachment, PVR, neurodegeneration; Gene therapy; cell therapy
Geographies of interest for exchanges (optional)	EU/Associated Countries; UK sociated Countries; UK; North America; Latin America; Asia

Overview of your research / innovation focus
Our team focuses on translational retinal research, with a particular emphasis on rhegmatogenous retinal detachment, proliferative vitreoretinopathy, neuroinflammation, photoreceptor degeneration and precision medicine in vitreoretinal surgery. Over the last years, our group has developed a clinically oriented research line linking genetic susceptibility, inflammatory pathways, retinal imaging biomarkers and functional outcomes after retinal detachment surgery. A central element of this work is the RAISE-VISION programme, which aims to move from conventional prognostic assessment towards personalised risk stratification based on molecular biomarkers, multimodal imaging and artificial intelligence. Our recent work has identified and validated the TP53 Arg72Pro polymorphism as a prognostic biomarker associated with visual and neurodegenerative outcomes after retinal detachment, supported by both human clinical data and humanised mouse models. The next translational step is to integrate rapid point-of-care genotyping, clinical and OCT/OCTA imaging variables into explainable AI-based decision-support tools. The ultimate goal is to identify high-risk patients early, support surgical prioritisation, guide intensified follow-up and enable genotype-informed neuroprotective strategies. The team combines clinical retina expertise, molecular and cellular biology, preclinical models, retinal organoids, imaging analysis, AI development and medical-device translation.

Why your team could be relevant to a translational eye-health and vision consortium
Our team could contribute strong added value to a translational eye-health and vision consortium because it bridges clinical ophthalmology, molecular biomarker discovery, preclinical validation and enabling technologies for real-world implementation. The group is embedded in a hospital–academic environment involving the Hospital Clínico Universitario de Valladolid, IOBA–University of Valladolid, IBioVALL and the RICORS-REI network, providing access to well-characterised retinal detachment cohorts, multimodal imaging, surgical expertise and translational laboratory platforms.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

• Clinical and translational expertise in vitreoretinal diseases, particularly rhegmatogenous retinal detachment, proliferative vitreoretinopathy, photoreceptor degeneration and functional recovery after retinal surgery.
• Longstanding experience in genetic risk stratification for PVR and poor visual outcomes, including TP53 Arg72Pro, TNF, TNFR2, SMAD7, PIK3, LTA, BAX and BCL-2-related pathways.
• Prospective clinical phenotyping of retinal detachment patients, including surgical variables, macular status, timing from symptoms, anatomical outcomes, PVR development and longitudinal functional recovery.
• Multimodal retinal imaging analysis, including OCT, OCT-A, autofluorescence, wide-field imaging and structural biomarkers such as photoreceptor integrity, ellipsoid zone/ELM status, LPOS, hyperreflective dots and cystoid macular changes.
• Molecular and cellular profiling of retinal detachment, including genotyping, transcriptomic analysis, cytokine/inflammatory profiling, apoptosis/autophagy markers and immune-cell characterisation.
• Preclinical validation using humanised TP53 Arg72Pro knock-in mouse models and retinal organoid/ex vivo systems to explore genotype-dependent neurodegeneration and neuroprotective strategies.
• Development of AI-based predictive models integrating clinical, imaging and genetic data for personalised risk stratification and clinical decision support.
• Translational development of point-of-care diagnostic tools, including Lab-on-a-Chip / rapid genotyping approaches designed for perioperative or early clinical use.
• Experience in clinical implementation, regulatory interaction, ethics submissions, MDR/IVDR-oriented device translation and health-technology adoption pathways.
• Capacity to connect clinical retina specialists, molecular scientists, engineers, AI developers, biotech partners and public healthcare networks within a single translational workflow.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Hospital-based vitreoretinal surgery setting at the Hospital Clínico Universitario de Valladolid, enabling prospective recruitment, clinical phenotyping and real-world validation of diagnostic or decision-support tools.

IOBA–University of Valladolid and IBioVALL translational research environment, combining clinical ophthalmology, molecular biology, imaging analysis and biomedical innovation.

Existing well-characterised retinal detachment cohort with clinical, imaging and genetic data, including the RICORS-Retina4 dataset used for AI model development and validation.

Access to multimodal ophthalmic imaging platforms, including OCT/OCT-A, fundus imaging, wide-field retinal imaging and image-based biomarker extraction.

Molecular and cellular biology laboratories for genotyping, inflammatory biomarker analysis, cytokine profiling, apoptosis/autophagy studies and validation of candidate biomarkers.

Biobank-supported management of biological samples, including blood, vitreous and retinal material obtained under approved ethical protocols.

Preclinical platforms including humanised TP53 Arg72Pro mouse models and retinal organoid/ex vivo models for mechanistic and therapeutic validation.

AI and data-science collaboration capacity through engineering and computer-science partners, including development of explainable machine-learning models and clinical dashboards.

Point-of-care diagnostic development capacity through collaboration with biotech and engineering partners for Lab-on-a-Chip, LAMP-based or immunochromatographic systems.

Regulatory, ethics and translational support through CEIm-approved clinical research, IBioVALL/UVa structures, OTRI support and previous experience in medical-device and software-as-medical-device translation.

Training capacity and knowledge-transfer potential

Inbound researchers, clinicians, engineers or innovation fellows could receive practical training in a highly translational ophthalmology environment, moving from patient care to biomarker discovery, preclinical validation and clinical implementation. Specifically, they could learn:
• Clinical pathways in retinal detachment and proliferative vitreoretinopathy, including surgical decision-making, timing of intervention and follow-up strategies.
• Standardised collection of clinical, imaging and biological data in prospective retinal cohorts.
• Interpretation of multimodal retinal imaging and extraction of clinically meaningful OCT/OCT-A biomarkers.
• Genetic and molecular biomarker workflows, including sample handling, genotyping strategies, inflammatory profiling and integration with clinical outcomes.
• Design and validation of AI-based prediction models for ophthalmology, including explainability, clinical usability and integration into decision-support workflows.
• Translational use of humanised animal models and retinal organoid/ex vivo systems to test genotype-dependent mechanisms and neuroprotective interventions.
• Development and early validation of point-of-care diagnostic devices for ophthalmic use.
• Practical aspects of ethics approval, GDPR-compliant data management, biobanking, MDR/IVDR-oriented device development and preparation for clinical translation.
• Knowledge transfer in project coordination, multicentre collaboration, public healthcare implementation, dissemination, guideline input and health-economic evaluation.

What your team would hope to gain from participating

Through participation in a translational eye-health and vision consortium, our team would aim to strengthen the international validation, scalability and clinical adoption of our precision-medicine approach in retinal detachment. We would particularly value:

- Access to complementary patient cohorts for external validation of genetic, imaging and AI-based risk prediction models.
- Collaboration with centres experienced in advanced retinal imaging, automated image segmentation, digital biomarkers and clinical AI deployment.
- Access to additional omics, single-cell, proteomic or spatial-biology platforms to deepen mechanistic understanding of neuroinflammation, apoptosis, fibrosis and photoreceptor degeneration.
- Collaboration with groups working on retinal organoids, advanced preclinical models, drug screening and neuroprotective or anti-inflammatory therapeutic strategies.
- Industrial and engineering expertise to accelerate the maturation of point-of-care genotyping devices, microfluidics, diagnostic assays and software-as-medical-device development.
- Support in regulatory strategy, IVDR/MDR compliance, quality-management systems, usability testing and health-technology assessment.
- Opportunities for joint publications, shared datasets, harmonised clinical protocols and multicentre grant applications.
- Exposure to broader translational pathways across Europe, including patient-oriented validation, implementation science, payer perspectives and public healthcare adoption.
- Long-term strategic collaboration to transform RAISE-VISION from a strong local/national translational programme into a scalable European precision-medicine platform for retinal disease.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
SALVADOR PASTOR IDOATE	Associate Prof. Dr.	Group name: Biomarkers in Neurodegenerative, Metabolic, and Inflammatory Pathology Department of Cell Biology. Faculty of Medicine. University of Valladolid IOBA- University of Valladolid			Salvador.pastor@uva.es
RICARDO USATEGUI MARTÍN	Prof.	Group name: Biomarkers in Neurodegenerative, Metabolic, and Inflammatory Pathology Department of Cell Biology. Faculty of Medicine. University of Valladolid IOBA- University of Valladolid			ricardo.usategui@uva.es

Main disciplines represented in your team	Ophthalmology and vitreoretinal surgery; translational retinal research; cell biology; molecular biology; genetics/genomics; immunology and neuroinflammation; retinal imaging and image analysis; preclinical models and retinal organoids; biomaterials and microfluidics; biomedical engineering; artificial intelligence/data science; clinical epidemiology and biostatistics; regulatory affairs, ethics and medical-device translation.
Experience in EU / international / multi-partner collaborative projects	The team has experience in multicentre and international retinal research through the Retina 4 Project, the RICORS-REI network, IBioVALL/IOBA-University of Valladolid collaborations, and formal partnerships with leading European ophthalmology centres in Spain, Italy, Germany and the United Kingdom. The group also participates in ERN-EYE-related activities and has coordinated translational projects combining clinical cohorts, genetic biomarkers, preclinical models, AI and point-of-care diagnostic development.
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
• Publications in peer-reviewed journals: 53. 22 in Q1 journals https://pubmed.ncbi.nlm.nih.gov/?term=pastor-idoate+S 66. 26 in Q1 journals. https://pubmed.ncbi.nlm.nih.gov/?term=usategui-martin+R • ORCID iD (SPI): 0000-0002-6463-4227 and (RUM): 0000-0001-7699-4388 • Web of Science Researcher ID (SPI): IAM-1368-2023. H-Index: 14; (RUM):): IAM: 56245265700 H-Index: 18 • Salvador Pastor: Two Research Six-Year periods (2012-2023) recognized by the National agency for Quality Assessment of Spain (ANECA). Ricardo Usategui: Two Research Six-Year periods by ANECA • The 5 most relevant publications: 1- The p53 codon 72 polymorphism (rs1042522) is associated with proliferative vitreoretinopathy: the Retina 4 Project. Ophthalmology. 2013 Mar;120(3):623-628. doi: 10.1016/j.ophtha.2012.08.019. Epub 2012 Dec 1. PMID: 23207172. 2- BAX and BCL-2 polymorphisms, as predictors of proliferative vitreoretinopathy development in patients suffering retinal detachment: the Retina 4 project. Acta Ophthalmol. 2015 Nov;93(7):e541-9. doi: 10.1111/aos.12718. Epub 2015 May 19. PMID: 25991504. 3- The T309G MDM2 gene polymorphism is a novel risk factor for proliferative vitreoretinopathy. PLoS One. 2013 Dec 9;8(12):e82283. doi: 10.1371/journal.pone.0082283. PMID: 24349246; PMCID: PMC3857251. 4- Proliferative vitreoretinopathy: A new concept of disease pathogenesis and practical consequences. Prog Retin Eye Res. 2016 Mar;51:125-55. doi: 10.1016/j.preteyeres.2015.07.005. Epub 2015 Jul 21. PMID: 26209346. 5- Galindo-Cabello N, Sobas-Abad EM, Lapresa R, Agulla J, Almeida Á, López A, Pastor JC, Pastor-Idoate S, Usategui-Martín R. The TP53 Arg72Pro polymorphism predicts visual and neurodegenerative outcomes in retinal detachment. Cell Death Dis. 2025 May 26;16(1):415. doi: 10.1038/s41419-025-07739-1. PMID: 40419469; PMCID: PMC12106684.

Up to 5 relevant previous projects, grants, networks or collaborations
Beneficiaries / participating investigators: Prof. Salvador Pastor-Idoate (SPI) and Prof.. Ricardo Usategui-Martín (RUM) Relevant funded projects and research networks: 1. PID2020-114585RA-I00 – National R&D project funded by the Spanish State Research Agency. 2. PID2023-147958OB-I00 – National R&D project funded by the Spanish State Research Agency. 3. RD21/0002/0017 – RICORS-REI Network – Related research groups: GR10, GR11, GR14, GR32 and GR41. 4. RD24/0001/0008 – RICORS-REI Network – Related research groups: GR10, GR11, GR14, GR32 and GR41. 5. PDC2023-145857-I00 / PDC2023-35 145857-100 – Proof-of-concept / translational innovation project. 6. CLU-2023-1-04 “Escalera de Excelencia” – Excellence programme supporting strategic research capacity and institutional strengthening.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
N/A

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation’s/team’s capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes X Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff X Other profiles: Clinical; Innovation / business; Data / software
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Original rough estimates provided: Sending capacity: 1–2 staff, depending on funding availability and clinical/teaching workload; approximately 1–4 person-months in total, preferably through short stays or split visits; short stays of 2–4 weeks; exceptionally 1–2 months if fully funded and compatible with clinical/academic duties; clinical researchers, clinician-scientists, early-career researchers, data/imaging collaborators, innovation-oriented staff. Hosting capacity: 1–3 incoming staff, depending on profile, supervision needs and available workspace; approximately 1–6 person-months in total, preferably through short stays or split visits; short stays of 2–8 weeks; flexible split stays would be preferred over long continuous stays; clinical researchers, preclinical/translational researchers, technical staff, data/software profiles, innovation/business development profiles. Incoming and outgoing mobility would need to be assessed case by case according to institutional rules of the University of Valladolid, Hospital Clínico Universitario de Valladolid, IOBA and/or IBioVALL, as applicable.

	Our team is open to staff exchange, but mobility should be planned as externally funded, short-term and clearly task-oriented, as the organisation cannot normally provide salary support for incoming or outgoing staff.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

Clinically embedded translational retina team with strong expertise in retinal detachment, proliferative vitreoretinopathy, vitreoretinal surgery, visual outcomes and real-world healthcare implementation.

Precision-medicine approach in retinal disease, integrating genetic biomarkers, neuroinflammation, photoreceptor degeneration, multimodal imaging and functional outcomes.

Validated biomarker and cohort assets, including experience with TP53 Arg72Pro and other genetic predictors of PVR/visual prognosis, supported by clinical cohorts, imaging data and preclinical validation.

AI and point-of-care innovation pipeline, focused on combining clinical variables, OCT/OCTA imaging and rapid genetic testing into explainable decision-support tools for early risk stratification.

Strong translational ecosystem, linking hospital care, IOBA–University of Valladolid, IBioVALL, RICORS-REI, preclinical models, organoid research, engineering/data-science partners and medical-device-oriented development.

Capabilities or expertise you would especially like to access through the consortium

We would particularly like to connect with partners offering complementary expertise in:

• External clinical validation cohorts in retinal detachment, proliferative vitreoretinopathy, inherited retinal disease or other neuroinflammatory retinal conditions.

• Advanced retinal imaging and digital biomarkers, including automated OCT/OCTA segmentation, image harmonisation, quantitative imaging pipelines and multimodal imaging repositories.

• AI, explainable machine learning and clinical decision-support deployment, especially teams experienced in trustworthy AI, federated learning, data interoperability, clinical dashboards and software-as-medical-device pathways.

• Point-of-care diagnostics and microfluidics, including Lab-on-a-Chip development, LAMP/qPCR miniaturisation, biosensors, assay validation, usability testing and IVDR-oriented diagnostic translation.

• Omics and spatial biology platforms, including single-cell transcriptomics, proteomics, metabolomics, spatial transcriptomics and immune-cell profiling in ocular tissues.

• Preclinical and organoid-based therapeutic testing, particularly teams working on retinal organoids, ex vivo retinal models, neuroprotection, anti-inflammatory therapies, anti-fibrotic strategies and drug screening.

• Regulatory, health-economic and implementation-science expertise, including MDR/IVDR strategy, quality management systems, health technology assessment, payer perspectives and adoption within public healthcare systems.

• Industry and innovation partners interested in co-developing scalable diagnostic, digital-health or precision-medicine tools for ophthalmology

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Overall, our team is strongly interested in participating in a translational eye-health and vision consortium, particularly where activities are clearly task-oriented, adequately funded, ethically approved and aligned with our clinical, scientific and innovation priorities in precision retinal medicine.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:

Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)